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951. Non-Euclidian Geometry.

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... then the values are

$$\lambda = rr_1\sqrt{1 - (M - M_1)^2}, \quad \mu = -rr_1\sqrt{1 - \dots}$$

And, this being so, the two systems of values of A, B, C, F, G, H , are

$$\begin{array}{ll} \lambda(2l + a) + \mu(2l - a), & \lambda(2l + a) - \mu(2l - a), \\ \lambda(S_3 + \beta) + \mu(S_3 - \beta), & \lambda(S_3 + \beta) - \mu(S_3 - \beta), \\ \lambda(S_2 + \gamma) + \mu(S_2 - \gamma), & \\ \lambda(S_1 + a) - \mu(S_1 - a), & \lambda(S_1 + a) + \mu(S_1 - a), \\ \lambda(S_3 + \beta) - \mu(S_3 - \beta), & \lambda(S_3 + \beta) + \mu(S_3 - \beta), \\ \lambda(S_2 + \gamma) + \mu(S_2 - \gamma), & \end{array}$$

viz. the two perpendiculars are reciprocals each of the other.

35. Before going further I notice that, if

$$\frac{a_1 + f_1}{a + f} = \frac{b_1 + g_1}{b + g} = \frac{c_1 + h_1}{c + h}, \quad \text{and} \quad \frac{a_1 - f_1}{a - f} = \frac{b_1 - g_1}{b - g} = \frac{c_1 - h_1}{c - h},$$

then the four equations for (A, B, C, F, G, H) reduce themselves to three equations only: and thus instead of two perpendiculars we have a singly infinite series of perpendiculars (see ante No. 15).

To explain the meaning of the equations, I observe that a line (a, b, c, f, g, h) will be a generating line of the one kind, or say a “generatrix,” of the Absolute if

$$a + f = 0, \quad b + g = 0, \quad c + h = 0;$$

and it will be a generating line of the other kind, or say a “directrix,” of the Absolute if $a - f = 0$, $b - g = 0$, $c - h = 0$. Or what is the same thing, we have

$$(a, b, c, -a, -b, -c), \text{ where } a^2 + b^2 + c^2 = 0 \text{ for a generatrix,}$$

and

$$(a, b, c, a, b, c), \text{ where } a^2 + b^2 + c^2 = 0 \text{ for a directrix, of the Absolute.}$$

Consider now two directrices (a, b, c, a, b, c) and $(a_1, b_1, c_1, a_1, b_1, c_1)$: if a line (a, b, c, f, g, h) meets each of these, then

$$\begin{aligned} (a + f)a + (b + g)b + (c + h)c &= 0, \\ (a + f)a_1 + (b + g)b_1 + (c + h)c_1 &= 0, \end{aligned}$$

and consequently

$$a + f : b + g : c + h = bc_1 - b_1c : ca_1 - c_1a : ab_1 - a_1b;$$

and similarly if $(a_1, b_1, c_1, f_1, g_1, h_1)$ meets each of the two directrices, then

$$a_1 + f_1 : b_1 + g_1 : c_1 + h_1 = bc_1 - b_1c : ca_1 - c_1a : ab_1 - a_1b,$$

that is, if the lines each of them meet the same two directrices of the Absolute, then

$$\frac{a_1 + f_1}{a + f} = \frac{b_1 + g_1}{b + g} = \frac{c_1 + h_1}{c + h}.$$

Conversely, if these relations are satisfied, then the lines each of them meet two directrices of the Absolute.

In like manner, if the lines each meet two generatrices of the Absolute, then

$$\frac{a_1 - f_1}{a - f} = \frac{b_1 - g_1}{b - g} = \frac{c_1 - h_1}{c - h};$$

and conversely, if these relations are satisfied, then the lines each of them meet the same two generatrices of the Absolute. In the former case, the lines are said to be “right parallels”: in the latter case, “left parallels.”

A line (a, b, c, f, g, h) meets the Absolute in two points, and through each of these we have a directrix and a generatrix: that is, the line meets two directrices and two generatrices.

Through a given point we may draw, meeting the two directrices, or meeting the two generatrices, a line: that is, through a given point we may draw a line

$$(a_1, b_1, c_1, f_1, g_1, h_1)$$

which is a right parallel, and a line which is a left parallel, to a given line. That is, regarding as given the first line, and also a point of the second line, there are two positions of the second line such that for each of them, the $\perp$ s of the pair of lines, instead of being two determinate lines, are a singly infinite series of lines.

36. Reverting to the general case, we have found (A, B, C, F, G, H) the coordinates of either of the lines $\perp$ to the given lines (a, b, c, f, g, h) and $(a_1, b_1, c_1, f_1, g_1, h_1)$: supposing that the $\perp$ intersects the first of these lines in the point the coordinates of which are (x, y, z, w) , and the second in the point the coordinates of which are (x_1, y_1, z_1, w_1) , then we have for each set of coordinates a fourfold expression; the choice of the form is indifferent, and I write

$$\begin{aligned} x : y : z : w &= cB - bC : aC - cA : bA - aB : fA + gB + hC, \\ x_1 : y_1 : z_1 : w_1 &= c_1B - b_1C : a_1C - c_1A : b_1A - a_1B : f_1A + g_1B + h_1C. \end{aligned}$$

We have then, for the distance of these two points,

$$\cos \theta = \dots$$

where $\phi = \delta$ or θ , according to the sign of the radical $\sqrt{\quad}$ contained in the expressions for A, B, C, F, G, H .

I have not succeeded in obtaining in this manner the final formulae for the determination of the distances: these in fact are, by what precedes, given by the equations

$$\sin \delta \sin \theta = M, \quad \cos \delta \cos \theta = M_1.$$

For then, writing ϕ to denote either of the distances δ, θ , at pleasure, we have

$$\sin^2 \phi \cos^2 \phi = MM_1,$$

that is,

$$\cos^4 \phi - \cos^2 \phi (M^2 - M_1^2 + 1) + M_1^2 = 0,$$

or

$$\cos^2 \phi = \frac{M^2 - M_1^2 + 1 \pm \sqrt{(M^2 - M_1^2 + 1)^2 - 4M_1^2}}{2},$$

which is the expression for the cosine of the distance.

In the case where the two lines intersect $M = 0$, and if δ be the $\perp$ distance which vanishes, then $\delta = 0$, and consequently $\cos \theta = M_1$: the last-mentioned formula, putting therein $M = 0$ and taking the radical to be $= M_1^2 - 1$, gives $\cos^2 \phi = M_1^2$, that is, $\phi = \theta$, and $\cos^2 \theta = M_1^2$, as it should be.

37. I verify as follows, in the case in question of two intersecting lines,

$$(af_1 + bg_1 + ch_1 + a_1f + b_1g + c_1h = 0),$$

the formula

$$\cos \theta = \frac{xx_1 + yy_1 + zz_1 + ww_1}{\sqrt{x^2 + y^2 + z^2 + w^2} \sqrt{x_1^2 + y_1^2 + z_1^2 + w_1^2}}.$$

We have here

$$A = S_1 = bc_1 - b_1c + fgh_1 - f_1g_1h,$$

$$B = S_2 = ca_1 - c_1a + hfg_1 - h_1f_1g,$$

$$C = S_3 = ab_1 - a_1b + ghf_1 - g_1h_1f,$$

$$F = \alpha = b_1h - bh_1 - cg_1 + c_1g,$$

$$G = \beta = c_1f - cf_1 - ah_1 + a_1h,$$

$$H = \gamma = a_1g - ag_1 - bf_1 + b_1f.$$

I stop to notice that these formulae may be obtained in a different and somewhat more simple manner: the two lines (a, b, c, f, g, h) and $(a_1, b_1, c_1, f_1, g_1, h_1)$ intersect; hence their reciprocals also intersect: the equation of the plane through the two lines and that of the plane through the two reciprocal lines are respectively

$$(g_1h - gh_1)x + (h_1f - hf_1)y + (f_1g - fg_1)z + (af_1 + bg_1 + ch_1)w = 0,$$

$$(bc_1 - b_1c)x + (ca_1 - c_1a)y + (ab_1 - a_1b)z + (fa_1 + gb_1 + hc_1)w = 0;$$

the line (A, B, C, F, G, H) is thus the line of intersection of these two planes, and it is thence easy to obtain the foregoing values.

From the values of A, B, C, F, G, H , we have to find x, y, z, w and x_1, y_1, z_1, w_1 by the formulae given above. We have

$$\begin{aligned} x &= cB - bC = bc(a_1 - a) + c(fg_1 - f_1g) + c(hf - h_1f) \\ &\quad - ab(b_1) + a_1b^2 - bf g_1 + \dots \\ &= (bg + ch)f_1 + (h(b^2 + c^2) - b_1hb - c_1hc - bf g_1 - cf h_1) \\ &= -f(af_1 + bg_1 + ch_1) + a_1(b^2 + c^2) - b_1hb - c_1hc; \end{aligned}$$

or writing here $af_1 + b_1g + c_1h$ in place of $-(af_1 + bg_1 + ch_1)$, this is a linear function of a_1, b_1, c_1 ; and similarly finding the values of y, z, w , we have

$$\begin{aligned} x &= a_1(b^2 + c^2 + f^2) + b_1(fg - hb) + c_1(hf - ca), \\ y &= a_1(fg - ab) + b_1(c^2 + a^2 + g^2) + c_1(gh - bc), \\ z &= a_1(hf - ca) + b_1(gh - bc) + c_1(a^2 + b^2 + h^2), \\ w &= a_1(bh - cg) + b_1(cf - ah) + c_1(ag - bf). \end{aligned}$$

And in like manner, (I introduce for convenience the sign $-$, as is allowable),

$$\begin{aligned} -x_1 &= a_1(b^2 + c^2 + f^2) + b_1(fg_1 - a_1h) + c_1(h_1f - c_1a), \\ -y_1 &= a_1(h_1f_1 - c_1g_1) + \dots, \\ -z_1 &= a_1(\dots) + \dots + c_1(a^2 + b^2 + h^2), \\ -w_1 &= a_1(bh_1 - c_1g_1) + b_1(c_1f - a_1h) + c_1(ag_1 - b_1f). \end{aligned}$$

38. Write for shortness

$$p = a^2 + b^2 + c^2, \quad p_1 = f^2 + g^2 + h^2, \quad q = af + bg + ch,$$

$$r = f^2 + g^2 + h^2, \quad r_1 = a^2 + b^2 + c^2, \quad \text{and therefore } q_1 = a_1f + b_1g + c_1h;$$

and also

$$\omega = af_1 + bg_1 + ch_1, \quad \omega_1 = a_1f + b_1g + c_1h.$$

We have

$$\begin{aligned} x &= cp - cq + h\omega, & x_1 &= -cr + dq + h_1\omega, \\ y &= \dots, & y_1 &= \dots, \\ z &= cp - cq + h\omega, & z_1 &= -cr + dq + h_1\omega, \\ w &= -c\dots, & w_1 &= -c\dots \end{aligned}$$

from which we easily obtain

$$\omega^2 + x^2 + z^2 = p(pr - q^2) + (p_1 + 2p)\omega^2,$$

and by expressing ω^2 in the form of a determinant

$$\omega^2 = p(pr - q^2) - p\omega^2,$$

we obtain

$$x^2 + y^2 + z^2 + w^2 = (r + r_1)(pr - q^2 + \omega^2),$$

and in like manner

$$x_1^2 + y_1^2 + z_1^2 + w_1^2 = (r + r_1)(pr - q^2 + \omega^2).$$

And again

$$xx_1 + yy_1 + zz_1 = q(pr - q^2) + (q_1 + 2q)\omega^2$$

and by expressing ww_1 in the form of a determinant

$$ww_1 = q_1(pr - q^2) - q\omega^2,$$

we find

$$xx_1 + yy_1 + zz_1 + ww_1 = (q + q_1)(pr - q^2 + \omega^2).$$

Hence substituting in

$$\cos \theta = \frac{xx_1 + yy_1 + zz_1 + ww_1}{\sqrt{x^2 + y^2 + z^2 + w^2} \sqrt{x_1^2 + y_1^2 + z_1^2 + w_1^2}}$$

the factor $pr - q^2 + \omega^2$ disappears, and we have

$$\cos \theta = \frac{q + q_1}{r + r_1},$$

the required result.

952. On the Kinematics of a Plane, and in particular on Three-bar Motion: and on a Curve-tracing Mechanism.

[From the Transactions of the Cambridge Philosophical Society, vol. xv. (1894), pp. 391—402.]

The first part of the present paper, On the Kinematics of a Plane, and on Three-bar Motion, is purely theoretical: the second part contains a brief description of a Curve-tracing Mechanism, which at my suggestion has been constructed by Prof. Ewing in the workshops of the Engineering Laboratory, Cambridge.

Part I.

1. The theory of the motion of a plane, when two given points thereof describe given curves, has been considered by Mr S. Roberts in his paper, “On the motion of a plane under given conditions,” Proc. Lond. Math. Soc. t. III. (1871), pp. 286—318, and he has shown that, if for the given curves the order, class, number of nodes, and of cusps, are (m, n, δ, κ) and $(m', n', \delta', \kappa')$ respectively ($n = m^2 - m - 2\delta - 3\kappa$, $n' = m'^2 - m' - 2\delta' - 3\kappa'$), then for the curve described by any fixed point of the plane:

$$\begin{aligned}\text{order} &= 2mm', \\ \text{class} &= 2(mm' + mn' + nm'), \\ \text{number of nodes} &= mm'(2mm' - m - m') + 2(m\delta' + m'\delta), \\ \text{number of cusps} &= 2(m\kappa' + m'\kappa);\end{aligned}$$

but he remarks that these formulae require modification when the directrices or either of them pass through the circular points at infinity. And he has considered the case where the two directrices become one and the same curve.

2. It will be convenient to speak of the line joining the two given points as the *link*; the two given points, say B and D , are then the extremities of the link; and I take the length of the link to be $= c$, and the two directrices to be b and d ; we have thus the link $c = BD$ moving in such wise that its extremity B describes the curve b of the order m , and its extremity D the curve d of the order m' : in Mr Roberts' problem, the locus is that described by a point P rigidly connected with the link, or say by a point P the vertex of the triangle PBD .

3. The points B, D describe of course the directrices b, d respectively: taking on b a point B_1 at pleasure, then if B be at B_1 , the corresponding positions of D are the intersections of d by the circle centre B_1 and radius c , viz. there are thus $2m'$ positions of D : and similarly taking on d a point D_1 at pleasure, then if D be at D_1 , the corresponding positions of B are the intersections of b by the circle centre D_1 and radius c , viz. there are thus $2m$ positions of B . The motion thus establishes a $(2m, 2m')$ correspondence between the points of the directrices b and d , viz. to a given point on b there correspond $2m'$ points on d , and to a given point on d there correspond $2m$ points on b . Of course, for a given position on either directrix, the corresponding points on the other directrix may be any or all of them imaginary; and thus it may very well be that for either directrix not the whole curve but only a part or detached parts thereof will be actually described in the course of the motion.

In saying that a part is described, we mean described by a continuous motion; say that the point B (the point D remaining always on a part of d) is capable of describing continuously a part of b ; it may very well happen that the point B (the point D remaining always on a different part of d) is capable of describing continuously a different part of b , but that it is not possible for B to pass from the one to the other of these parts of b without removing D from the one

part and placing it on the other part of d , and thus that we have on b detached parts each of them continuously described by B ; and similarly, we may have on d detached parts each of them continuously described by D .

4. But dropping for the moment the question of reality, to a given position of B on b there correspond as was mentioned $2m'$ positions of D on d , or say $2m'$ positions of the link c : in the entire motion of the link it must assume each of these $2m'$ positions, and for each of them the point B comes to assume the position in question on b ; the directrix b is thus described $2m'$ times, that is, the locus described by B will be the directrix b repeated $2m'$ times, or say a curve of the order $m \times 2m', = 2mm'$. Similarly, the locus described by D will be the directrix d repeated $2m$ times, or say a curve of the order $m' \times 2m, = 2mm'$.

5. In general, if B_1D_1 be any position of the link and if B moves from B_1 along b in a determinate sense, then D will move from D_1 along d in a determinate sense; and if B moves from B_1 along b in the opposite sense, then also D will move from D_1 along d in the opposite sense. Or what is the same thing, we may have B moving in a determinate sense through B_1 , and D moving in a determinate sense through D_1 ; and reversing the sense of B 's motion, we reverse also the sense of D 's motion. But there are certain critical positions of the link, viz. we have a critical position when

the link is a normal at B_1 to the directrix b , or a normal at D_1 to the directrix d .

Say first the link is a normal at B_1 to the directrix b . The infinitesimal element at B_1 may be regarded as a straight line at right angles to the link; hence if for a moment D_1 is regarded as a fixed point, the link may rotate in either direction round D_1 , that is, B may move from B_1 along b in either of the two opposite senses, say B_1 is a “two-way point.” But if on d we take on opposite sides of D_1 the consecutive points D'_1 and D''_1 , say $D'_1 D_1$ cuts $D_1 B_1$ at an acute angle and $D''_1 D_1$ cuts it at an obtuse angle, then D'_1 will be nearer to b than was D_1 , and thus the circle centre D'_1 and radius c will cut b in two real points B'_1 and B''_1 near to and on opposite sides of B_1 ; or as D moves to D'_1 , B will move from B_1 indifferently to B'_1 or B''_1 . Contrariwise, D''_1 is further from b than was D_1 , and thus the circle centre D''_1 and radius c , will not meet b in any real point near to B_1 , and hence D is incapable of moving from D_1 in the sense $D_1 D''_1$. Or what is the same thing, the described portion of d , which includes a point D'_1 , will terminate at D_1 , or say D_1 is a “summit” on the directrix d . We have thus a summit on d , corresponding to the two-way point on b . And of course in like manner, if the link is a normal at D_1 to the directrix d , then D_1 is a two-way point on d , and the corresponding point B_1 is a summit on b .

6. If the link is at the same time a normal at B_1 to b and at D_1 to d , then each of the points B_1 , D_1 is a two-way point and also a summit; or more accurately, each of them is a two-way point and also a pair of coincident summits.

But the case requires further investigation. Considering the position $B_1 D_1$ as given, we may take the axis of x coincident with this line, and the origin O in such wise that OB_1 , OD_1 are each positive and $OD_1 > OB_1$; say we have $OD_1 = \delta$, $OB_1 = \beta$, and therefore $\delta - \beta = c$. The equation of the curve b in the neighbourhood of B_1 is $y^2 = 2p(x - \beta)$, where p is the radius of curvature at B_1 , assumed to be positive when the curve is convex to O , or what is the same thing when the centre of curvature R lies to the right of B_1 ($OR - OB_1 = +$); and similarly the equation of d in the neighbourhood of D_1 is $y^2 = 2\sigma(x - \delta)$, where σ is the radius of curvature at D_1 assumed to be positive when the curve is convex to O , or what is the same thing when the centre of curvature S lies to the right of D_1 ($OS - OD_1 = +$).

Consider now (x_1, y_1) the coordinates of a point on b in the neighbourhood of B_1 , $y_1^2 = 2p(x_1 - \beta)$, and taking B at this point, let (x_2, y_2) be the coordinates of the corresponding point D on d in the neighbourhood of D_1 , $y_2^2 = 2\sigma(x_2 - \delta)$. We have

$$c^2 = (x_1 - x_2)^2 + (y_1 - y_2)^2,$$

and here

$$\begin{aligned} x_1 - x_2 &= \frac{y_1^2}{2p} - \frac{y_2^2}{2\sigma} + (\delta - \beta) = \frac{y_1^2}{2p} - \frac{y_2^2}{2\sigma} + c, \\ (y_1 - y_2)^2 &= y_1^2 - 2y_1 y_2 + y_2^2, \end{aligned}$$

whence

$$\left(\frac{y_1^2}{2p} - \frac{y_2^2}{2\sigma}\right)^2 + 2c\left(\frac{y_1^2}{2p} - \frac{y_2^2}{2\sigma}\right) + (y_1 - y_2)^2 = 0.$$

The equation thus becomes

$$\frac{1}{4}\left(\frac{y_1^2}{p} - \frac{y_2^2}{\sigma}\right)^2 + c\left(\frac{y_1^2}{p} - \frac{y_2^2}{\sigma}\right) + (y_1 - y_2)^2 = 0,$$

that is,

$$\frac{y_1^2}{p}\left(\frac{y_1^2}{4p} + c\right) + \frac{y_2^2}{\sigma}\left(\frac{y_2^2}{4\sigma} - c\right) - \frac{y_1^2 y_2^2}{2p\sigma} - 2y_1 y_2 + y_1^2 + y_2^2 = 0,$$

a quadric equation between y_1 and y_2 . Evidently if we had taken D a point on d , coordinates (x_2, y_2) , in the neighbourhood of D_1 and had sought for the coordinates (x_1, y_1) of the corresponding point B on b in the neighbourhood of B_1 , we should have found the same equation between y_1 and y_2 .

7. The equation will have real roots if

$$p\sigma > (p + \beta - \delta)(\sigma + \delta - \beta),$$

viz. p, σ having the same sign, this is

$$p\sigma > (p + \beta - \delta)(\sigma + \delta - \beta) :$$

but p, σ having opposite signs, then

$$p\sigma < (p + \beta - \delta)(\sigma + \delta - \beta).$$

These conditions may be written

$$(OR - OB_1)(OS - OD_1) - (OS - OB_1)(OR - OD_1) > \text{ or } < 0,$$

that is,

$$(OS - OR)(OD_1 - OB_1) > \text{ or } < 0.$$

But we have $OD_1 - OB_1 = +$, and therefore, p, σ having the same sign, the condition of reality is $OS > OR$, i.e. S to the right of R ; but p, σ having opposite signs, the condition of reality is $OS < OR$, i.e. S to the left of R . Observe that, S lying to the left of R , we cannot have $p = -$, $\sigma = +$, and that the second alternative thus is $p = +$, $\sigma = -$, then $OS < OR$, or S lies to the left of R .

The condition was investigated as above in order to exhibit more clearly the geometrical signification: but of course the original form, or say the equation

$$\frac{1}{p\sigma} \left[1 - \left(1 + \frac{\beta - \delta}{p} \right) \left(1 + \frac{\delta - \beta}{\sigma} \right) \right] > 0$$

gives at once

$$1 - \left(1 + \frac{\beta - \delta}{p} \right) \left(1 + \frac{\delta - \beta}{\sigma} \right) > 0.$$

8. Writing the quadric equation in the form

$$ay_1^2 + 2hy_1y_2 + by_2^2 = 0,$$

we have

$$a = \frac{1}{p} \left(\frac{y_1^2}{4p} + c \right) + 1, \quad b = \frac{1}{\sigma} \left(\frac{y_2^2}{4\sigma} - c \right) + 1, \quad 2h = -\frac{y_1y_2}{p\sigma} - 2;$$

the two values of $y_1 : y_2$ will have the same sign or opposite signs according as $1 - \frac{c}{p}$ and $1 + \frac{c}{\sigma}$ have the same sign or opposite signs, and in the case where these have the same sign, then this is also the sign of each of the two values of $y_1 : y_2$.

Or what is the same thing, if $1 - \frac{c}{p}$ and $1 + \frac{c}{\sigma}$ are each of them positive, then the two values of $y_1 : y_2$ are each of them positive; if $1 - \frac{c}{p}$ and $1 + \frac{c}{\sigma}$ are each of them negative, then the two values of $y_1 : y_2$ are each of them negative; and if $1 - \frac{c}{p}$ and $1 + \frac{c}{\sigma}$ have opposite signs, then the two values of $y_1 : y_2$ have opposite signs. Considering the different cases $p, \sigma = +, +; +, -; -, +; -, -$, we find

$p, \sigma = +, +$, then values of $y_1 : y_2$ are $++$ or $--$, according as DR, DS are $++$ or $--$;

$p, \sigma = -, -$, then values of $y_1 : y_2$ are $++$ or $--$, according as DR, DS are $--$ or $++$;

$p, \sigma = +, -$, then values of $y_1 : y_2$ are $+-$, according as DR, DS are $+-$ or $-+$;

$p, \sigma = -, +$, then values of $y_1 : y_2$ are $+-$, according as DR, DS are $-+$ or $+-$;

and in each case the values of $y_1 : y_2$ are $++$ or $--$, if the two distances referred to have opposite signs: $DR = +$ means that R is to the right of, or beyond, D , and so in other cases.

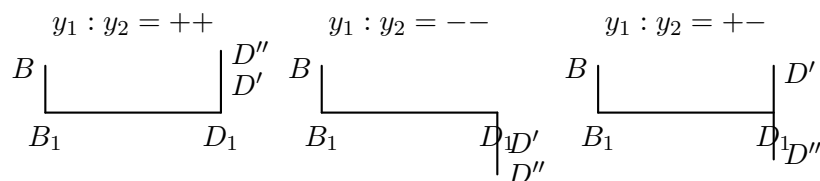
9. The different cases, two real roots as above, are

	$y_1 : y_2$	
$p, \sigma = ++$	$++$	$R \quad S$ $\delta \quad \beta$
$p, \sigma = --$	$--$	$S \quad R$ $\beta \quad \delta$
$p, \sigma = +-$	$+-$	$\delta \quad R \quad S \quad \beta$
$p, \sigma = -+$	$+-$	$\beta \quad S \quad R \quad \delta$

Obviously the cases $p, \sigma = --$ correspond exactly to the cases $p, \sigma = ++$: the only difference is that the concavities, instead of the convexities, of the two curves are turned towards the point O .

10. If the two roots of the quadratic equation are imaginary, then B_1 is a conjugate or isolated position of the link, and B_1, D_1 are isolated points on the curves b and d respectively.

11. If the roots are real, then the three cases $y_1 : y_2 = ++, --$, and $+-$, may be delineated as in the annexed figures, viz. taking in each case y_1 as positive, that is, imagining B to move upwards from B_1 through an infinitesimal arc of b , then D moves from D_1 through either of two infinitesimal arcs of d , both upwards, both downwards, or the one upwards and the other downwards, as shown in the figures.



and where it is to be observed that, reversing the sense of the motion of B from B_1 , we reverse also the senses of the motion of D from D_1 : moreover that, considering D as moving through an infinitesimal arc of d from D_1 , we have the like relations thereto of the two infinitesimal arcs of b described by B from B_1 . Thus the points B_1 and D_1 are singular points of like character.

If $y_1 : y_2 = ++$, we may say that B_1 (or D_1) is a *for-forwards point*; if $y_1 : y_2 = --$, then that B_1 (or D_1) is a *back-backwards point*; and if $y_1 : y_2 = +-$, then that B_1 (or D_1) is a *back-forwards point*.

12. The separating case between two imaginary roots and two real roots is that of two equal roots, viz. the condition is

$$p\sigma = (p + \beta - \delta)(\sigma + \delta - \beta),$$

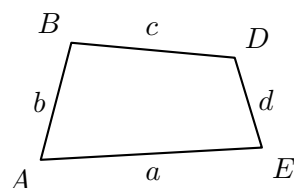
or, as this may be written,

$$\frac{1}{p} + \frac{1}{\sigma} = \frac{1}{\delta - \beta} = \frac{1}{c},$$

a condition independent of the aberrancies of curvature of the curves b and d at these points respectively. If each of the curves is a circle, then the curves are concentric circles, and the link BD moves in suchwise that its direction passes always through the common centre of the two circles—or say so that BD is always a radius of the annulus formed by the two circles—and

the foregoing condition is satisfied identically for every position of the link.

13. I consider in particular the case where the directrices b, d are circles. The mechanism is then the “three-bar motion” of Mr Roberts; viz. the link $c = BD$ has its extremities to the arms or radii $AB = b$, and $ED = d$, which rotate about the fixed centres A, E at a distance from each other $= a$. Here a, b, c, d are each of them positive; a, b, d may have any values, but then c is at most $= a + b + d$, and if $a > b + d$ then c is at least $= a - b - d$; but if $a =$ or $< b + d$, then c may be $= 0$, viz. it may have any value from 0 to $a + b + d$. And in either case there will be critical values of c . The cases are very numerous. To make an exhaustive enumeration, we may assume d at most $= b$, and in each of the two cases $d < b$ and $d = b$, considering the centre of the circle d as moving from the right of the centre of the circle b towards this centre, we may in the first instance divide as follows:



$d = b$	$d < b$
$\odot d$ exterior to $\odot b$, touches it externally, cuts it, touches it internally, is concentric and thus coincident with it;	$\odot d$ exterior to $\odot b$, touches it externally, cuts it, touches it internally, lies within it, is concentric with it,

and then, in each of these cases, give to the length c of the link its different admissible values.

14. Considering the case $d < b$, then we have (see Plate I., at p. 516), exterior series, the figures 1, 1–2, 2, 2–3, 3, 3–4, 4, viz.

- fig. 1, $c = a - b - d$,
- fig. 1–2, intermediate,
- fig. 2, $c = a - b + d$,
- fig. 2–3, intermediate,
- fig. 3, $c = a + b - d$,
- fig. 3–4, intermediate,
- fig. 4, $c = a + b + d$.

15. In figure 1, the curves described by the extremities B and D respectively are each of them a mere point.

In figure 1–2, we have $a + d > b + c$ and $a + b > d + c$. Hence in the course of the motion the arms b, c come into a right line, giving a position B_1D_1 of the link, where B_1 is a two-way point on b and D_1 a summit on d ; or rather, there are two such positions symmetrically situate on opposite sides of the axis Ax . And again, in the course of the motion, the arms d, c come into a right line, giving a position $B'_1D'_1$, where D'_1 is a two-way point on d and B'_1 a summit on b ; or rather, there are two such positions symmetrically situate on opposite sides of the axis Ax . Only an arc of the circle b is described, viz. the arc adjacent to d included between the two summits B'_1 on b ; and in like manner, only an arc of the circle d is described, viz. the arc

adjacent to b included between the two summits B'_1 on d . The described portions on b and d respectively are to be regarded each of them as a double line or indefinitely thin bent oval: and it is to be observed that for a given position of B (or D) there are two positions of the link BD , each of these positions being assumed by the link in the course of its motion.

16. In figure 2, the two positions B_1D_1 of the link come to coincide together in a single axial position BD , but we still have the other two positions $B'_1D'_1$ of the link, where B'_1 is a summit on b , and D'_1 a two-way point on d . As regards BD , this is the configuration $p, \sigma = +-, R, \delta; y_1 : y_2 = \pm$, and thus each of the axial points B, D is a back-and-forwards point. Thus only the arc B'_1B_1 of the circle b is described by the point B , but the whole circumference of the circle d is described by the point D . If we further examine the motion it will appear that, as B moves from the axial point B say to the upper summit B_1 and returns to B , then D starting from the axial point D may describe (and that in either sense, viz. $y_1 = +$, then we have $y_2 = \pm$) the entire circumference of d , returning to the axial point D ; and similarly, as B moves from the axial point B to the lower summit B'_1 and returns to B , then D starting as before from the axial point D may describe (and that in either sense, viz. $y_1 = -$, then we have $y_2 = +$) the entire circumference of d , returning to the axial point D .

17. In figure 2–3, we have still the two positions B_1D_1 symmetrically situate on opposite sides of the axis, B_1 a two-way point on b , D_1 a summit on d ; and also the two positions $B'_1D'_1$ symmetrically situate on opposite sides of the axis, B'_1 a summit on b , D'_1 a two-way point on d . The described portion of b is the arc B'_1B_1 , and the described portion of d is the arc D'_1D_1 , these arcs being thus the nearer portions of the two circles respectively. But we have moreover a new pair of critical positions B_1D_1 situate symmetrically on opposite sides of the axis: the configuration is $p, \sigma = +- , B, R, S, D$; and the values of $y_1 : y_2$ are $+-$; that is, B_1 (or D_1) is a back-forwards point: the described portion of b is still the arc B'_1B_1 , but that of d is, as before, the entire circumference. It is to be observed (as in fig. 2) that properly it is not the entire arc B'_1B_1 , but each of the half-arcs BB_1, BB'_1 , which corresponds to the entire circumference of d .

18. In figure 3, the two positions $B'_1D'_1$ come to coincide in the single axial position BD ; we have thus only the two positions B_1D_1 symmetrically situate on opposite sides of the axis; the configuration is $p, \sigma = ++ , B, D, R, S$; $y_1 : y_2 = ++$. The described portion of b is the entire circumference, and that of d is, as before, the entire circumference. It is to be observed (as in fig. 2) that properly it is not the entire arc B'_1B_1 , but each of the half-arcs BB_1, BB'_1 , which corresponds to the entire circumference of d .

19. The figure 3–4 closely corresponds to fig. 1–2, the only difference being that the arcs B_1B_1 and $D'_1D'_1$, which are the described portions of b and d respectively (instead of being the nearer portions, or those with their convexities facing each other) are the further portions, or those with their concavities facing each other, of the two circles respectively.

Finally, in fig. 4, the described portions of the two circles reduce themselves to the axial points B and D respectively.

20. Still assuming $d < b$, and passing over the case of external contact, we come to that in which the circles intersect each other; but this case has to be subdivided. Since the circles intersect, we have $b + d > a$, consistently herewith we may have:

- b, d each $< a$, A, E each outside the lens common to the two circles,
- $b = a, d < a$, A outside, E on boundary of the lens,
- $b > a, d < a$, A outside, E inside the lens,
- $b > a, d = a$, A on boundary of, E inside the lens,
- b, d each $> a$, A, E , each inside the lens;

and in each case we have to consider the different admissible values of c . I omit the discussion of all these cases.

21. Still assuming $d < b$, and passing over the case of internal contact, we come to that of the circle d included within the circle b : we have here again a subdivision of cases; viz. we may have $d > a$, that is, A inside d ; $d = a$, that is, A on the circumference of d ; or $d < a$, that is, A outside d . The critical values of c , arranged in order of increasing magnitude in these three cases respectively, are:

$d > a$	$d = a$	$d < a$
$b - 2d,$		
$b - d - a,$	$b - d - a,$	
$b + d - a,$	$b + d - a,$	$b - d - a,$
$b - d + a,$	$b - d + a,$	$b + d - a,$
$b + d + a,$	$b + 2d,$	$b - d + a,$
		$b + d + a.$

I attend only to the first case; we have here (see Plate II., at p. 516), interior series, the figures 1, 1–2, 2, 2–3, 3, 3–4, 4.

22. In figure 1, the curves described by the points B , D are each of them a mere point. In figure 1–2, we have two critical positions B_1D_1 situate symmetrically on opposite sides of the axis, B_1 being a summit on b , and D_1 a two-way point on d , and moreover two critical positions $B'_1D'_1$ situate symmetrically on opposite sides of the axis, B'_1 being a two-way point on b , and D'_1 a summit on d . The described portion of b is the arc B'_1B_1 , and the described portion of d is the arc D'_1D_1 , these two arcs being thus the nearer portions of the two circles respectively.

23. In figure 2, the four critical positions coalesce all of them in the axial position BD ; the described portions are thus the entire circumferences of the two circles respectively. This is a remarkable case. The configuration is p , $\sigma = ++$; B , D , R , S ; $y_1 : y_2 = ++$. Imagine D to move from the axial point D in a given sense round the circle d , say with uniform velocity, then B moves from the axial point B in the same sense but with either of two velocities round the circle b ; one of these velocities is at first small but ultimately increases rapidly, the other is at first large but ultimately decreases rapidly, so that the two revolutions of B from the axial point B round the entire circumference to the axial point B correspond each of them to the revolution of D from the axial point D round the entire circumference to the axial point D . And similarly, if we imagine B to move in a given sense from the axial point B round the circle b , say with uniform velocity, then D moves from the axial point D in the same sense but with either of two velocities round the circle d : one of these velocities is at first small but ultimately increases rapidly, the other is at first large but ultimately decreases rapidly, so that the two revolutions from the axial point D round the entire circumference of d to the axial point D correspond each of them to the revolution of B from the axial point B round the entire circumference of b to the axial point B .

24. In figure 2–3, the four critical positions again exist, viz. we have the two positions B_1D_1 symmetrically situate on opposite sides of the axis, B_1 a two-way point on b , D_1 a summit on d ; and also the two positions $B'_1D'_1$ symmetrically situate on opposite sides of the axis, B'_1 a summit on b , D'_1 a two-way point on d . The described portion of b is the arc B'_1B_1 , and the described portion of d is the arc D'_1D_1 , these arcs being thus the nearer portions of the two circles respectively. But we have moreover, as in fig. 2–3 of the exterior series, a new pair of critical positions B_1D_1 situate symmetrically on opposite sides of the axis; the configuration is $p, \sigma = +-, B, R, S, D$; and the values of $y_1 : y_2$ are $+-$; the described portion of b is still the arc B'_1B_1 , but that of d is, as in fig. 2, the entire circumference.

25. In figure 3, the two positions $B'_1D'_1$ again coalesce in the axial position BD ; we have thus only the two positions B_1D_1 symmetrically situate on opposite sides of the axis; the configuration is $p, \sigma = ++, B, D, R, S$; $y_1 : y_2 = ++$. The described portion of b is the entire circumference, and that of d is, as in fig. 2, the entire circumference; but observe that, if we imagine B starting from the axial point B in a given sense round the circle b say with uniform velocity, then D moves from the axial point D round the circle d in the opposite sense, and with either of two velocities.

26. In figure 3–4, we have again the two critical positions B_1D_1 symmetrically situate on opposite sides of the axis, B_1 a summit on b , D_1 a two-way point on d ; and also the two critical positions $B'_1D'_1$ symmetrically situate on opposite sides of the axis, B'_1 a two-way point on b , D'_1 a summit on d . The described portion of b is the arc B'_1B_1 , and the described portion of d the arc D'_1D_1 , these arcs being thus the further portions of the two circles respectively.

Finally, in figure 4, the described portions reduce themselves to the two points B, D respectively.

27. The several forms for $d = b$ can be at once obtained from those for $d < b$; the only difference is that several intermediate forms disappear, and the entire series of divisions is thus not quite so numerous.

Part II.

1. The curve-tracing mechanism was devised with special reference to the curves of three-bar motion, viz. the object proposed was that of tracing the curve described by a point K of the link BD , the extremities whereof B and D describe given circles respectively, or more generally by a point K , the vertex of a triangle KBD , whereof the other vertices B and D describe given circles respectively, and that in suchwise that the points B and D might be free to describe the two entire circumferences respectively: but the principle applies to other motions, and I explain it in a general way as follows.

2. Imagine the cranked link BD , composed of the bars $B\beta$ and $D\delta$, rigidly attached $B\beta$ to the top and $D\delta$ to the bottom of the cylindrical disk K (this same letter K is used to denote the axis of the disk), and where $B\beta$ and $D\delta$ may be either parallel or inclined to each other at any given angle, so that, referring the points B, K, D to a horizontal plane, BKD is either a right line, or else K is the vertex of a triangle the other vertices whereof are B and D . The disk K , with the attached bars $B\beta$ and $D\delta$, moves in a horizontal plane: and if the motion of the point B be regulated in any manner by a mechanism lying wholly below B and supported by the bed of the entire mechanism, and similarly if the motion of the point D be regulated in any manner by a mechanism lying wholly above D and supported by a bridge of sufficient length (resting on the bed of the entire mechanism), then the disk K moves in its own horizontal plane unimpeded by other parts of the mechanism: and if we fit the disk K so as to move smoothly

within a circular aperture in the arm of a pentagraph, then the pencil of the pentagraph will trace out on a sheet of paper the curve described by the point K on the axis of the disk, or say by the point K of the beam BKD . Of course for the three-bar motion, all that is required is that the point B shall describe a circle, viz. it must be pivoted on to an arm AB , which is itself pivoted at A to the bed: and that the point D shall describe a circle, viz. it must be pivoted on to an arm DE , which is itself pivoted at E to the bridge.

Special arrangements are required to enable the variation of the several lengths AB , BK , KD , DE and ED , and the mechanism thus unavoidably assumes a form which appears complicated for the object intended to be thereby effected.

3. The form of Pentagraph which I use consists of a parallelogram $ABCD$, pivoted together at the points A , B , C , D , the bars AB and DC being above AD and BC . There is a cradle G , rotating about a fixed centre, and which carries between guides the arm AD , which has a sliding motion, so that the lengths GD and GA may be made to have any given ratio to each other. Above the bar DC and sliding along it, we have the arm KL (where K is the circular aperture which fits on to the disk K of the cranked link): and above AB and sliding along it, we have the arm MP which carries the pencil P : of course, in order that the pentagraph may be in adjustment, the points K , G , P must be in line.

953. On the Nine-Points Circle.

[From the Messenger of Mathematics, vol. xxIII. (1894), pp. 23—25.]

If from the angles A, B, C of a triangle we draw tangents to a conic H , meeting the opposite sides in the points $\alpha, \alpha'; \beta, \beta'; \gamma, \gamma'$ respectively, then it is known that these six points lie in a conic. In particular, if the conic H reduce itself to a point-pair OO' , then we have the theorem that, if from the angles A, B, C , we draw to the point O lines meeting the opposite sides in the points α, β, γ respectively; and to the point O' lines meeting the opposite sides in the points α', β', γ' respectively, then the six points $\alpha, \alpha'; \beta, \beta'; \gamma, \gamma'$ lie in a conic.

We may enquire the conditions under which this conic becomes a circle. It may be remarked that one of the points say O' remains arbitrary: for if through the points α', β', γ' , we draw a conic (or in particular a circle) meeting the three sides respectively in the remaining points α, β, γ , then (by a converse of the general theorem) the lines $A\alpha, B\beta, C\gamma$ will meet in a point O .

Using trilinear coordinates (x, y, z) and writing $x : y : z = a : b : c$ for the point O , and $x : y : z = a' : b' : c'$ for the point O' , it is at once seen that the equation of the conic through the six points is

$$aa'x^2 + bb'y^2 + cc'z^2 - (bc' + b'c)yz - (ca' + c'a)zx - (ab' + a'b)xy = 0;$$

in fact, writing herein successively $x = 0, y = 0, z = 0$, we see that the equation is satisfied by $x = 0, (by - cz)(b'y - c'z) = 0$; by $y = 0, (cz - ax)(c'z - a'x) = 0$; and by $z = 0, (ax - by)(a'x - b'y) = 0$ respectively. And it is to be observed that the equation may also be written

$$(aa'x + bb'y + cc'z)(x + y + z) - (b + c)(b' + c')yz - (c + a)(c' + a')zx - (a + b)(a' + b')xy = 0.$$

Suppose now that x, y, z represent areal coordinates, viz. that (x, y, z) are proportional to the perpendicular distances of the point from the sides, each divided by the perpendicular distance of the opposite angle from the same side; or, what is the same thing, coordinates such that the equation of the line infinity is $x + y + z = 0$. Then if A, B, C denote the angles of the triangle, the general equation of a circle is

$$(yz \sin^2 A + zx \sin^2 B + xy \sin^2 C) + (\lambda x + \mu y + \nu z)(x + y + z) = 0,$$

where λ, μ, ν are arbitrary coefficients.

Hence, putting this

$$= \Theta \{ -(b + c)(b' + c')yz - (c + a)(c' + a')zx - (a + b)(a' + b')xy + (aa'x + bb'y + cc'z)(x + y + z) \},$$

we must have

$$\begin{aligned} \Theta(b + c)(b' + c') &= -\sin^2 A, \\ \Theta(c + a)(c' + a') &= -\sin^2 B, \\ \Theta(a + b)(a' + b') &= -\sin^2 C; \end{aligned}$$

and then

$$\Theta aa' = \lambda, \quad \Theta bb' = \mu, \quad \Theta cc' = \nu,$$

which last equations determine the values of λ, μ, ν .

Taking a', b', c' at pleasure, we have

$$\begin{aligned}\frac{1}{\Theta} &= -\frac{\sin^2 A}{(b+c)(b'+c')} = -\frac{\sin^2 B}{(c+a)(c'+a')} = -\frac{\sin^2 C}{(a+b)(a'+b')}, \\ a &= \frac{1}{\Theta} \left(\frac{\sin^2 A}{b'+c'} - \frac{\sin^2 B}{c'+a'} - \frac{\sin^2 C}{a'+b'} \right), \\ b &= \frac{1}{\Theta} \left(-\frac{\sin^2 A}{b'+c'} + \frac{\sin^2 B}{c'+a'} - \frac{\sin^2 C}{a'+b'} \right), \\ c &= \frac{1}{\Theta} \left(-\frac{\sin^2 A}{b'+c'} - \frac{\sin^2 B}{c'+a'} + \frac{\sin^2 C}{a'+b'} \right); \end{aligned}$$

viz. a, b, c having these values, the conic through the six points $\alpha, \beta, \gamma, \alpha', \beta', \gamma'$ is the circle having for its equation

$$yz \sin^2 A + zx \sin^2 B + xy \sin^2 C + \Theta(aa'x + bb'y + cc'z)(x + y + z) = 0;$$

and we may obviously without loss of generality give to Θ any specific value, say $\Theta = 1$.

If $a' = b' = c' = 1$, then we have

$$\begin{aligned}-4a &= (-\sin^2 A + \sin^2 B + \sin^2 C), \\ -4b &= (\sin^2 A - \sin^2 B + \sin^2 C), \\ -4c &= (\sin^2 A + \sin^2 B - \sin^2 C), \end{aligned}$$

or writing for convenience $\Theta = -\frac{1}{4}$, the values of a, b, c are

$$\frac{1}{4}(-\sin^2 A + \sin^2 B + \sin^2 C), \quad \frac{1}{4}(\sin^2 A - \sin^2 B + \sin^2 C), \quad \frac{1}{4}(\sin^2 A + \sin^2 B - \sin^2 C)$$

respectively. But we have

$$A + B + C = \pi,$$

and thence

$$\begin{aligned}\sin^2 A + \sin^2 B - \sin^2 C &= \sin^2 A + \sin^2 B - \sin^2(A + B) \\ &= -2 \sin A \sin B (\sin A \sin B - \cos A \cos B), \\ &= -2 \sin A \sin B \cos(A + B), \\ &= 2 \sin A \sin B \cos C, \end{aligned}$$

and we thus have

$$a, b, c = \sin B \sin C \cos A, \sin C \sin A \cos B, \sin A \sin B \cos C,$$

(or, what is the same thing, $a : b : c = \cot A : \cot B : \cot C$), and the equation of the circle is

$$\begin{aligned}yz \sin^2 A + zx \sin^2 B + xy \sin^2 C \\ - \frac{1}{4}(x \sin B \sin C \cos A + y \sin C \sin A \cos B + z \sin A \sin B \cos C)(x + y + z) = 0.\end{aligned}$$

We thus have $x : y : z = 1 : 1 : 1$ for the point O' , and $x : y : z = \cot A : \cot B : \cot C$ for the point O ; viz. O' is the point of intersection of the lines from the angles to the mid-points of the opposite sides respectively; and O is the point of intersection of the perpendiculars from the angles on the opposite sides respectively: and the foregoing equation is consequently that of the *Nine-points Circle*.

954. On the Nine-Points Circle of a Plane Triangle.

[From the Messenger of Mathematics, vol. xxIII. (1894), pp. 25—27.]

I consider the circle which meets the sides of a triangle ABC in the points F, L, G, M, H, N respectively, where ultimately F, G, H are the feet of the perpendiculars let fall from the angles on the opposite sides, and L, M, N are the mid-points of the sides: but in the first instance, they are taken to be arbitrary points.

Taking the radius of the circle to be unity, the coordinates of the point F may be taken to be $\cos F, \sin F$, and these may be expressed rationally in terms of the tangent of the half-angle, $f = \tan \frac{1}{2}F$; and similarly for the other points, viz. we may determine the six points by the parameters f, g, h, l, m, n respectively. The sides of the triangle are the lines joining the points $L, F; M, G; N, H$ respectively: thus the equations of the sides are

$$\begin{aligned} \text{for } BC : \quad & x(1 - lf) + y(l + f) - (1 + lf) = 0, \quad \text{say } U = 0, \\ \text{" } CA : \quad & x(1 - mg) + y(m + g) - (1 + mg) = 0, \quad \text{" } V = 0, \\ \text{" } AB : \quad & x(1 - nh) + y(n + h) - (1 + nh) = 0, \quad \text{" } W = 0. \end{aligned}$$

We have AF , a line through the intersections of BC and CA ; its equation is therefore of the form $BV - CW = 0$, and to determine B, C we have $BV_0 - CW_0 = 0$, if V_0, W_0 are the values of V, W belonging to the point F , the coordinates of which are

$$\frac{1 - f^2}{1 + f^2}, \quad \frac{2f}{1 + f^2};$$

we find

$$V_0 = -2(f - g)(f - m), \quad W_0 = 2(h - f)(f - n);$$

and then $B : C = W_0 : V_0$; we thus find the following equations:

$$\begin{aligned} \text{that of } AF \text{ is } & BV - CW = 0, \\ \text{" } BG \text{ " } & C'W - A'U = 0, \\ \text{" } CH \text{ " } & A''U - B''V = 0, \end{aligned}$$

where

$$\begin{aligned} B : C &= -(h - f)(f - n) : (f - g)(f - m), \\ C' : A' &= -(f - g)(g - l) : (g - h)(g - n), \\ A'' : B'' &= -(g - h)(h - m) : (h - f)(h - l). \end{aligned}$$

The condition in order that the three lines may meet in a point is $BC'A'' = CA'B''$, viz. this is

$$(f - n)(g - l)(h - m) + (f - m)(g - n)(h - l) = 0,$$

or, as this may also be written,

$$fgh - gh(m + n) - hf(n + l) - fg(l + m) + mn(g + h) + nl(h + f) + lm(f + g) - 2lmn = 0.$$

Similarly, the equation of

$$\begin{aligned} AL \text{ is } & \mathfrak{B}V - \mathfrak{C}W = 0, \\ BM \text{ " } & \mathfrak{C}'W - \mathfrak{A}'U = 0, \\ CN \text{ " } & \mathfrak{A}''U - \mathfrak{B}''V = 0, \end{aligned}$$

where

$$\begin{aligned}\mathfrak{B} : \mathfrak{C} &= -(n-l)(h-l) : (l-m)(g-l), \\ \mathfrak{C}' : \mathfrak{A}' &= -(l-m)(f-m) : (m-n)(h-m), \\ \mathfrak{A}'' : \mathfrak{B}'' &= -(m-n)(g-n) : (n-l)(f-n).\end{aligned}$$

The condition in order that the three lines may meet in a point is $\mathfrak{B}\mathfrak{C}'\mathfrak{A}'' = \mathfrak{C}\mathfrak{A}'\mathfrak{B}''$, viz. this is the same condition as before; that is, if the lines AF , BG , CH meet in a point, then also the lines AL , BM , CN will meet in a point.

In the case of the nine-points circle, we have MN , NL , LM parallel to LF , MG , NH , respectively: the equation of MN is

$$x(1-mn) + y(m+n) - (1+mn) = 0,$$

and this is parallel to LF , if

$$\frac{1-mn}{1-lf} = \frac{m+n}{l+f}, \quad \text{i.e. if } m+n-l-mnf = 0.$$

Hence, for the nine-points circle, we have

$$l + F = M + N, \quad M + G = N + L,$$

or, as these equations may be written,

$$l = \frac{1}{2}(M + N + F - G), \quad m = \frac{1}{2}(N + L + G - H), \quad n = \frac{1}{2}(L + M + H - F);$$

viz. it thus appears that the radii to the points L , M , N respectively, or say the radii L , M , N , bisect the angles made by the radii G and H , H and F , F and G respectively.

It may be added that we have

$$\begin{aligned}m + n - l + lmn &= f\{1 - mn + l(m+n)\}, \\ n + l - m + lmn &= g\{1 - nl + m(n+l)\}, \\ l + m - n + lmn &= h\{1 - lm + n(l+m)\},\end{aligned}$$

viz. f , g , h are expressible each of them as a rational function of l , m , n .

955. The Numerical Value of $\Pi_i = \Gamma(1 + i)$.

[From the Messenger of Mathematics, vol. xxIII. (1894), pp. 36—38.]

I do not know whether the numerical value of Π_x for an imaginary value of x has ever been calculated; and I wish to calculate it for a simple case $x = i$.

We have

$$\Pi_i = \Gamma(1 + i) = i\Gamma(i) = i \int_0^\infty e^{-t} t^{i-1} dt;$$

and therefore, writing $t = e^{-\theta}$, $dt = -e^{-\theta} d\theta$,

$$\begin{aligned} \Pi_i &= i \int_{-\infty}^0 e^{e^{-\theta}} e^{-\theta(i-1)} (-e^{-\theta} d\theta) \\ &= i \int_0^\infty e^{-e^{-\theta}} e^{-i\theta} d\theta. \end{aligned}$$

Writing $e^{-e^{-\theta}} = \phi_0 - \phi_1\theta + \phi_2\theta^2 - \&c.$ (where $\phi_0 = e^{-1}$, and the law of the coefficients will be considered presently), we have

$$\Pi_i = i \int_0^\infty (\phi_0 - \phi_1\theta + \phi_2\theta^2 - \&c.) (\cos \theta - i \sin \theta) d\theta;$$

and observing that

$$\int_0^\infty \theta^n \cos \theta d\theta = n! \cos \frac{1}{2}n\pi, \quad \int_0^\infty \theta^n \sin \theta d\theta = n! \sin \frac{1}{2}n\pi,$$

we have

$$\Pi_i = i\{(\phi_0 - 1!\phi_2 - 3!\phi_4 + 5!\phi_6 + \&c.) + i(1!\phi_1 - 2!\phi_3 - 4!\phi_5 + 6!\phi_7 + \&c.)\},$$

($\phi_2 = 0$, and in the subsequent terms the imaginary part is taken with a negative sign in order to obtain positive values for $\phi_2, \phi_4, \&c.$), $= R(\cos \Theta + i \sin \Theta)$, if R be the modulus and Θ the amplitude.

We have

$$R^2 = (\phi_0 - 1!\phi_2 - 3!\phi_4 + \&c.)^2 + (1!\phi_1 - 2!\phi_3 - 4!\phi_5 + \&c.)^2,$$

which may be calculated directly. The value of Θ admits, however, of a finite expression.

956. On Richelot's Integral of the Differential Equation $\frac{dx}{\sqrt{X}} + \frac{dy}{\sqrt{Y}} = 0$.

[From the Messenger of Mathematics, vol. xxIII. (1894), pp. 42—47.]

In the Memoir “Einige neue Integralgleichungen des Jacobischen Systems Differentialgleichungen,” Crelle, t. xxv. (1843), pp. 97—118, Richelot, working with the more general differential equation

$$\frac{dx}{\sqrt{X}} + \frac{dy}{\sqrt{Y}} + \frac{dz}{\sqrt{Z}} = 0,$$

where X, Y, Z are quartic functions of x, y, z respectively, obtains the integral of this equation in a form involving the quotients of the two different hyperelliptic integrals belonging to the equation. And he remarks that the result may be applied to the integral of the equation

$$\frac{dx}{\sqrt{X}} + \frac{dy}{\sqrt{Y}} = 0,$$

where X, Y are each of them a quartic function of x, y respectively; in fact, writing herein $z = \text{constant}$, and therefore $dz = 0$, the equation becomes the equation in question. But the result is thereby presented in a form which is not immediately intelligible: and in the memoir “Sur l'intégration de l'équation

$$\frac{dx}{\sqrt{X}} + \frac{dy}{\sqrt{Y}} = 0,$$

où X, Y sont les mêmes fonctions de x et de y respectivement,” Crelle, t. c. (1887), pp. 290—294, I obtained the integral under a more simple form. I propose to reproduce the investigation, with some explanations which were omitted in the original memoir.

1. If in the differential equation

$$\frac{dx}{\sqrt{X}} + \frac{dy}{\sqrt{Y}} = 0,$$

we write $x = \frac{a\theta^2 + 2b\theta + c}{a'\theta^2 + 2b'\theta + c'}$, then y will be the same function of ϕ , where ϕ is determined by the equation

$$\frac{d\theta}{\sqrt{\Theta}} + \frac{d\phi}{\sqrt{\Phi}} = 0;$$

and we thus arrive at the standard form belonging to the differential equation in question.

Working with the standard form, if θ, ϕ are coordinates, the differential equation determines a family of curves; taking any curve of the family, we may from any given point of this curve determine a new point thereof by means of the integral equation; and the integral equation is in fact an equation between the parameters θ, ϕ belonging to the two points respectively.

2. I use the accentuate letters a, b, c, f, g, h to denote the coefficients of the sextic function

$$(a, b, c, f, g, h \mid \theta, 1)^6,$$

or, what is the same thing, the coefficients of the quartic function $(a, b, c, f, g, h \mid x, y)^4$, where $x : y = \theta : 1$.

The differential equation is

$$\frac{d\theta}{\sqrt{\Theta}} + \frac{d\phi}{\sqrt{\Phi}} = 0,$$

where $\Theta = (a, b, c, f, g, h \mid \theta, 1)^6$, and Φ is the same function of ϕ .

3. I remark that, starting from the differential equation, and for the moment using u instead of θ , so that the equation is

$$\frac{du}{\sqrt{U}} + \frac{dv}{\sqrt{V}} = 0,$$

then from the first equation we have $v = \text{const.}$, and from the second equation we have $u = \text{const.}$, and it thus appears that the integral of the differential equation may be written in the form $F(u, v) = 0$, where $F(u, v)$ is a symmetrical function of u, v . But the equation may also be written in the form $v = f(u)$, where f denotes a functional symbol: and from the symmetry we have $u = f(v)$, that is, $f(f(u)) = u$, or f is a periodic function of the second order.

957. Illustrations of Sylow's Theorems on Groups.

[From the Messenger of Mathematics, vol. xxIII. (1894), pp. 59—62.]

The theorems 1, 2, and 3 in the paper Sylow, "Theoremes sur les groupes de Substitutions," Math. Ann. t. V. (1872), pp. 584—594, apply to groups in general, and not only to groups of substitutions. They are as follows:

THEOREM 1. If n^α be the highest power of the prime number n which divides the order of a group G , this group contains a group g of the order n^α : if, moreover, $n^\alpha\nu$ is the order of the highest group contained in G , the operations whereof are permutable with the group g , then the order of G is of the form $n^\alpha\nu(nk + 1)$.

(I write k for Sylow's p , since it is convenient to have p to denote a prime number; and for Sylow's "Substitutions" I write "Operations.")

THEOREM 2. Everything being as in the preceding theorem, the group G contains precisely $nk + 1$ distinct groups of the order n^α ; and these are obtained by transforming any one of them by the operations of G , each group being given by $n^\alpha\nu$ distinct transformations.

THEOREM 3. If the order of a group is n^α , n being prime, then any operation S whatever of the group may be expressed by the formula

$$S = G_0^{\theta_0} G_1^{\theta_1} G_2^{\theta_2} \dots G_{\alpha-1}^{\theta_{\alpha-1}},$$

where

$$G_\lambda^{\alpha} = G_0^{a_0} G_1^{a_1} \dots G_{\lambda-1}^{a_{\lambda-1}},$$

and where

$$G_\mu G_\lambda = G_0^{c_0} G_1^{c_1} \dots G_\lambda^{c_\lambda} \cdot G_\lambda \cdot G_0^{d_0} G_1^{d_1} \dots G_{\mu-1}^{d_{\mu-1}}.$$

But at present I attend only to the theorems 1 and 2.

For instance, consider the group G of the order $N = 6$,

$$1, \beta, \beta^2, \alpha, \alpha\beta, \alpha\beta^2 \quad (\alpha^2 = 1, \beta^3 = 1, \alpha\beta^2 = \beta\alpha, \alpha\beta = \beta^2\alpha).$$

Here $n = 2$ or 3 : if $n = 2$, we have $N = n^\alpha\nu(nk + 1) = 2 \cdot 1 \cdot (2 + 1)$; if $n = 3$, we have $N = n^\alpha\nu(nk + 1) = 3 \cdot 2 \cdot 1$.

First, $n = 2$; we should have a group g of the order 2; one such group is $(1, \alpha)$, and the only group the substitutions whereof are permutable with $(1, \alpha)$ is the group $(1, \alpha)$ itself: for, taking any other operation of the group, for instance β , it is not true that $\beta(1, \alpha) = (1, \alpha)\beta$; in fact, the left-hand is $(\beta, \beta\alpha)$ and the right-hand is $(\beta, \alpha\beta)$ or $(\beta, \beta^2\alpha)$: hence $n^\alpha\nu = 2\nu = 2$, or $\nu = 1$.

Hence also, by theorem 2, there should be 3 groups of the order 2 such as $(1, \alpha)$, viz. these are $(1, \alpha)$, $(1, \alpha\beta)$, $(1, \alpha\beta^2)$, derived from $(1, \alpha)$ as follows:

$$\begin{aligned} 1 \cdot (1, \alpha) \cdot 1^{-1} &= (1, \alpha), \\ \alpha \cdot (1, \alpha) \cdot \alpha^{-1} &= (1, \alpha), \\ \beta \cdot (1, \alpha) \cdot \beta^{-1} &= (1, \alpha\beta), \\ \beta^2 \cdot (1, \alpha) \cdot \beta^{-2} &= (1, \alpha\beta^2), \\ \alpha\beta \cdot (1, \alpha) \cdot (\alpha\beta)^{-1} &= (1, \alpha\beta^2), \\ \alpha\beta^2 \cdot (1, \alpha) \cdot (\alpha\beta^2)^{-1} &= (1, \alpha\beta), \end{aligned}$$

since $\beta^{-1} = \beta^2$, and therefore the second term is $\beta^2\alpha\beta^2 = \alpha\beta^2 \cdot \beta^2 = \alpha\beta$,

$\beta^{-2} = \beta$, and therefore the second term is $\beta\alpha\beta = \alpha\beta \cdot \beta = \alpha\beta^2$,

$(\alpha\beta)^{-1} = \alpha\beta$, and therefore the second term is $\alpha\beta\alpha\alpha\beta = \alpha\beta^2\alpha\alpha\beta = \alpha\beta$,

$(\alpha\beta^2)^{-1} = \alpha\beta^2$, and therefore the second term is $\alpha\beta^2\alpha\alpha\beta^2 = \alpha\beta^2\alpha\alpha\beta^2 = \alpha\beta$;

viz. the derivatives are $(1, \alpha)$, $(1, \alpha\beta)$, $(1, \alpha\beta^2)$, each twice.

Secondly, $n = 3$; there should be here a group of the order 3, viz. this is $(1, \beta, \beta^2)$. The group, the substitutions whereof are permutable with $(1, \beta, \beta^2)$, is the entire group $(1, \beta, \beta^2, \alpha, \alpha\beta, \alpha\beta^2)$; in fact, taking any substitution thereof, for instance α , we have $\alpha(1, \beta, \beta^2) = (1, \beta, \beta^2)\alpha$, viz. the left-hand side is $(\alpha, \alpha\beta, \alpha\beta^2)$, and the right-hand side is $(\alpha, \beta\alpha, \beta^2\alpha) = (\alpha, \alpha\beta^2, \alpha\beta)$, which is the left-hand side, the change of order being immaterial; this is the meaning of the expression used, “the operations whereof are permutable with the group g .” Hence, we have $n^\alpha\nu = 3\nu = 6$, or $\nu = 2$; and thence also $nk + 1 = 3k + 1 = 1$, viz. $k = 0$. There is thus only a single group of the order 3, viz. the group $(1, \beta, \beta^2)$.

As another instance, I take the group of the order 12 formed by the positive substitutions of four letters, viz. these are

$$1, \quad ab \cdot cd, \quad ac \cdot bd, \quad ad \cdot bc, \\ abc, \quad acb, \quad abd, \quad adb, \quad acd, \quad adc, \quad bcd, \quad bdc.$$

Here, taking $n = 2$, we have $N = n^\alpha\nu(nk + 1) = 2^2 \cdot 3 \cdot 1$; there is a group g of the order 4, viz. this is

$$(1, ab \cdot cd, ac \cdot bd, ad \cdot bc),$$

and the greatest group, the substitutions whereof are permutable with this group g , is the entire group of the order 12; thus, considering any substitution thereof, for instance abc , we have

$$abc \left| \begin{array}{cccc} 1 & ab \cdot cd & ac \cdot bd & ad \cdot bc \\ abc & acd & bdc & adb \end{array} \right| = \left| \begin{array}{cccc} 1 & ab \cdot cd & ac \cdot bd & ad \cdot bc \\ abc & bdc & adb & acd \end{array} \right| abc,$$

viz. the left-hand is (abc, acd, bdc, adb) , the right-hand is (abc, bdc, adb, acd) ; hence $n^\alpha\nu = 4\nu = 12$ or $\nu = 3$; whence also $nk + 1 = 2k + 1 = 1$: and thus the foregoing group g is the only group of the order 4.

Similarly, taking $n = 3$, we have $N = n^\alpha\nu(nk + 1) = 3 \cdot 1 \cdot 4$. There is a group g of the order 3, say $(1, abc, acb)$; the greatest group, the substitutions whereof are permutable with g , is the group g itself, viz. we have $n^\alpha\nu = 3\nu = 3$, or $\nu = 1$; and then $n^\alpha(k + 1) = 3k + 1 = 4$: there are thus 4 groups of the order 3, viz. these are

$$(1, abc, acb), \quad (1, abd, adb), \quad (1, acd, adc), \quad (1, bcd, bdc).$$

Reverting to the before-mentioned group of the order 6, this not only contains each of the groups $(1, \alpha)$, $(1, \alpha\beta)$, $(1, \alpha\beta^2)$ of order 2, and the group $(1, \beta, \beta^2)$ of order 3, but it is also the permutable product of any one of the groups $(1, \alpha)$, $(1, \alpha\beta)$, $(1, \alpha\beta^2)$ of the order 2 by the group $(1, \beta, \beta^2)$ of the order 3; but it is the permutable product of a group of order 2 by a group of order 3, say it is

$$G = (1, \alpha)(1, \beta, \beta^2) = (1, \beta, \beta^2)(1, \alpha).$$

A group, which is thus a permutable product of two factors, is said to be a *true* product.

958. On the Surface of the Order n which passes through a given Cubic Curve.

[From the Messenger of Mathematics, vol. xxIII. (1894), pp. 79, 80.]

We may through a given cubic curve draw a surface of any given order n ; such surface depends upon a certain number of constants, and the question arises to find the number of the constants. As to this, observe that the number of constants is equal to that of the constants in the equation of the surface of the order n , diminished by the number of the conditions in order that the surface may pass through the given cubic curve. I propose to show that the number of conditions is $= 3n + 1$.

We may through any $3n + 1$ points of the cubic curve draw a surface of the order n ; this surface meets the cubic curve in $3n$ points, and therefore in one additional point; or, what is the same thing, the surface contains the cubic curve.

Taking the cubic curve to be the intersection of the quadric surface $y^2 - zx = 0$ and the cubic surface $yw - z^2 = 0$, we may write

$$x : y : z : w = \theta^3 : \theta^2 : \theta : 1,$$

and then any surface of the order n meets the cubic curve in $3n$ points, determined by an equation of the degree $3n$ in θ ; hence if the surface pass through $3n + 1$ points of the cubic curve, it passes through the whole curve.

The number of conditions thus is $= 3n + 1$; and the number of constants in the surface of the order n is $= \frac{1}{6}(n + 1)(n + 2)(n + 3)$; whence the number of the constants in the surface is

$$\frac{1}{6}(n + 1)(n + 2)(n + 3) - (3n + 1).$$

959. Note on Plucker's Equations.

[From the Messenger of Mathematics, vol. xxIV. (1895), pp. 23, 24.]

It is well known that if

$$A, B, C, D = 2, 3, 6, 8,$$

then the equations

$$\begin{aligned} n &= m^2 - m - A\delta - B\kappa, \\ \iota &= 3m^2 - 6m - C\delta - D\kappa, \\ m &= n^2 - n - A\iota - B\tau, \\ \kappa &= 3n^2 - 6n - C\iota - D\tau \end{aligned}$$

imply

$$\frac{1}{2}(m-1)(m-2) - \delta - \kappa = \frac{1}{2}(n-1)(n-2) - \iota - \tau,$$

viz. this last equation is the only relation independent of A, B, C, D which exists between the six quantities.

If A, B, C, D are arbitrary, the foregoing equations give

$$\begin{aligned} n - m^2 + m &= -A\delta - B\kappa, \\ 3m^2 - 6m - \iota &= C\delta + D\kappa, \end{aligned}$$

and thence eliminating κ , we have

$$B(3m^2 - 6m - \iota) - D(n - m^2 + m) = (BC - AD)\delta,$$

or say

$$\delta = \frac{B(3m^2 - 6m - \iota) - D(n - m^2 + m)}{BC - AD}.$$

Similarly, the same two equations eliminating δ give

$$\kappa = \frac{C(n - m^2 + m) - A(3m^2 - 6m - \iota)}{BC - AD};$$

and the equations involving ι, τ give in like manner

$$\begin{aligned} \iota &= \frac{B(3n^2 - 6n - \kappa) - D(m - n^2 + n)}{BC - AD}, \\ \tau &= \frac{C(m - n^2 + n) - A(3n^2 - 6n - \kappa)}{BC - AD}. \end{aligned}$$

But the relation $\frac{1}{2}(m-1)(m-2) - \delta - \kappa = \frac{1}{2}(n-1)(n-2) - \iota - \tau$ must be satisfied identically; and substituting the foregoing values of $\delta, \kappa, \iota, \tau$, the equation becomes identically true.

960. On the Circle of Curvature at any point of an Ellipse.

[From the Messenger of Mathematics, vol. xxIV. (1895), pp. 47, 48.]

Let

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1.$$

The equation of the circle of curvature at the point (x, y) of the ellipse is

$$X^2 + Y^2 + \frac{2}{a^2b^2}(-b^2Xu^2 + a^2Yv^2) + u^2(a^2 - 2b^2) + v^2(b^2 - 2a^2) = 0,$$

where

$$u^2 = \frac{x^2}{a^4} + \frac{y^2}{b^4}, \quad u = \frac{x}{a^2}, \quad v = \frac{y}{b^2}.$$

Writing $X = x + \xi$, $Y = y + \eta$, and observing that

$$\frac{x\xi}{a^2} + \frac{y\eta}{b^2} = 0,$$

the equation becomes

$$\xi^2 + \eta^2 + \frac{2}{a^2b^2}(-b^2(x + \xi)u^2 + a^2(y + \eta)v^2) + \text{terms independent of } \xi, \eta = 0,$$

and hence reducing by means of the foregoing relations, we find the equation of the circle of curvature in the form

$$\xi^2 + \eta^2 - \frac{2(a^2 - b^2)}{a^2b^2} \left(\frac{b^2x}{a^2}\xi + \frac{a^2y}{b^2}\eta \right) \frac{1}{u^2} = 0.$$

Or, what is the same thing, the circle of curvature at the point (x, y) has for its equation

$$(X - x)^2 + (Y - y)^2 = \frac{2(a^2 - b^2)}{a^2b^2} \left(\frac{b^2x}{a^2}(X - x) + \frac{a^2y}{b^2}(Y - y) \right) \frac{1}{u^2}.$$

The centre is given by

$$X = x - \frac{(a^2 - b^2)x}{a^4u^2}, \quad Y = y - \frac{(a^2 - b^2)y}{b^4u^2},$$

and the radius of curvature is

$$\rho = \frac{(a^2 - b^2)^{3/2}}{a^2b^2} \cdot \frac{xy}{u^3}.$$

961. A Trigonometrical Identity.

[From the Messenger of Mathematics, vol. xxIV. (1895), pp. 49—51.]

The following was proposed as a Senate House Problem: Given the equations

$$\begin{aligned} a \cos(\beta + \gamma) + b \cos(\beta - \gamma) + c &= 0, \\ a \cos(\gamma + \alpha) + b \cos(\gamma - \alpha) + c &= 0, \\ a \cos(\alpha + \beta) + b \cos(\alpha - \beta) + c &= 0, \end{aligned}$$

to eliminate α, β, γ , and find the relation between a, b, c .

Writing the equations in the forms

$$\begin{aligned} (a + b) \cos \beta \cos \gamma + (-a + b) \sin \beta \sin \gamma + c &= 0, \\ (a + b) \cos \gamma \cos \alpha + (-a + b) \sin \gamma \sin \alpha + c &= 0, \\ (a + b) \cos \alpha \cos \beta + (-a + b) \sin \alpha \sin \beta + c &= 0, \end{aligned}$$

and writing for shortness

$$p = a + b, \quad q = -a + b, \quad r = c,$$

the equations become

$$\begin{aligned} p \cos \beta \cos \gamma + q \sin \beta \sin \gamma + r &= 0, \\ p \cos \gamma \cos \alpha + q \sin \gamma \sin \alpha + r &= 0, \\ p \cos \alpha \cos \beta + q \sin \alpha \sin \beta + r &= 0. \end{aligned}$$

We hence find

$$\begin{aligned} \cos \alpha : \sin \alpha : 1 &= (q \sin \beta \sin \gamma + r) : -(p \cos \beta \cos \gamma + r) : (p \cos \beta \sin \gamma - q \sin \beta \cos \gamma), \\ \cos \beta : \sin \beta : 1 &= (q \sin \gamma \sin \alpha + r) : -(p \cos \gamma \cos \alpha + r) : (p \cos \gamma \sin \alpha - q \sin \gamma \cos \alpha), \\ \cos \gamma : \sin \gamma : 1 &= (q \sin \alpha \sin \beta + r) : -(p \cos \alpha \cos \beta + r) : (p \cos \alpha \sin \beta - q \sin \alpha \cos \beta). \end{aligned}$$

Writing $x = \tan \frac{1}{2}\alpha$, $y = \tan \frac{1}{2}\beta$, $z = \tan \frac{1}{2}\gamma$, we have

$$\cos \alpha = \frac{1 - x^2}{1 + x^2}, \quad \sin \alpha = \frac{2x}{1 + x^2}, \quad \cos \beta = \frac{1 - y^2}{1 + y^2}, \quad \sin \beta = \frac{2y}{1 + y^2}, \quad \cos \gamma = \frac{1 - z^2}{1 + z^2}, \quad \sin \gamma = \frac{2z}{1 + z^2}.$$

Substituting these values, the first equation becomes

$$p(1 - x^2)(1 - y^2)(1 - z^2) + 4qxyz + r(1 + x^2)(1 + y^2)(1 + z^2) = 0,$$

and similarly the other two equations may be obtained by cyclical interchanges of x, y, z .

The elimination of x, y, z from these three equations leads to the required relation between a, b, c .

We have $a, b, 2c$ equal to the values given by the determinant; and then, substituting in the first equation, we have

$$x(1 + 2y^2z^2)(x^2 - yz) + x(y^2 + z^2)(-1 + x^2yz) + (1 - x^2)(yz^2 + yz^2) = 0,$$

which is

$$(x - y)(x - z)[x + y + z - (yz + zx + xy)xyz] = 0,$$

viz. our relation between x, y, z is

$$x + y + z - (yz + zx + xy)xyz = 0,$$

or, what is the same thing,

$$\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = xyz\left(\frac{1}{yz} + \frac{1}{zx} + \frac{1}{xy}\right).$$

The elimination hence gives

$$a^2 - b^2 + 2bc = 0,$$

which is the required relation.

The equation to be proved may also be written

$$\begin{vmatrix} \cos(\beta + \gamma) & \cos(\beta - \gamma) & 1 \\ \cos(\gamma + \alpha) & \cos(\gamma - \alpha) & 1 \\ \cos(\alpha + \beta) & \cos(\alpha - \beta) & 1 \end{vmatrix} = 0,$$

which is at once seen to be true.

962. Coordinates versus Quaternions.

[From the Proceedings of the Royal Society of Edinburgh, vol. xx. (1895), pp. 271—275. Read July 2, 1894.]

It is contended that Quaternions (as a method) are more comprehensive and less laborious than Cartesian Coordinates. It is not easy to meet this contention by a formal reply; but it is very easy to show that, as a matter of fact, in the solution of geometrical problems, the method of Coordinates is more powerful and more convenient than that of Quaternions.

The question was raised in the Society in connexion with a theorem, “Given the edges of a tetrahedron, find the radius of the circumscribed sphere.” I gave a solution by coordinates; and Professor Tait, while admitting that the coordinates were competent to deal with the problem, gave as his view, that they were inappropriate to it. I do not understand the appropriateness or inappropriateness of a method; the question is, which method gives the more easily the result required.

The quaternion solution depends on the identity

$$V\alpha\beta\gamma = S\alpha\beta\gamma S\alpha\beta - S\alpha\gamma S\beta + S\beta\gamma S\alpha,$$

and the condition, in order that the vectors α, β, γ may be coplanar, is $S\alpha\beta\gamma = 0$, and Professor Tait contrasts this with the prolixity of the corresponding coordinate formula.

But the vector formula $V\alpha\beta\gamma = \dots$ is not the formula which should be used in the solution of the problem. The proper formula is

$$\alpha^2\beta^2\gamma^2 = (S\beta\gamma)^2\alpha^2 + (S\gamma\alpha)^2\beta^2 + (S\alpha\beta)^2\gamma^2 - 2S\beta\gamma S\gamma\alpha S\alpha\beta,$$

which, when $S\alpha\beta\gamma = 0$, reduces to

$$0 = (S\beta\gamma)^2\alpha^2 + (S\gamma\alpha)^2\beta^2 + (S\alpha\beta)^2\gamma^2 - 2S\beta\gamma S\gamma\alpha S\alpha\beta.$$

This is the formula which should be used; and when translated into coordinates, it gives at once the required result.

I would further remark that the quaternion formula, even when it is the most appropriate, requires to be translated into coordinates in order to be understood. As an illustration, I compare a quaternion formula to a pocket-map—a capital thing to put in one’s pocket, but which for use must be unfolded: the formula, to be understood, must be translated into coordinates.

I remark that the imaginary of ordinary algebra—for distinction call this θ —has no relation whatever to the quaternion symbols i, j, k ; in fact, in the general quaternion $w + ix + jy + kz$, the coefficients w, x, y, z are, and must be, ordinary real or imaginary magnitudes.

And for the comparison of the two solutions, we have

$$\alpha = i(x_2 - x_1) + j(y_2 - y_1), \quad \beta = i(x_4 - x_3) + j(y_4 - y_3).$$

But this example of a plane theorem is a trivial one, given only for the sake of completeness.

963. Note on Dr Muir's Paper, "A Problem of Sylvester's in Elimination."

[From the Proceedings of the Royal Society of Edinburgh, vol. xx. (1895), pp. 306—308.]

The problem is, given the equations

$$\begin{aligned} U &= ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy = 0, \\ V &= a'x^2 + b'y^2 + c'z^2 + 2f'yz + 2g'zx + 2h'xy = 0, \\ W &= a''x^2 + b''y^2 + c''z^2 + 2f''yz + 2g''zx + 2h''xy = 0, \end{aligned}$$

to eliminate (x, y, z) , and obtain the condition for the coexistence of the three equations.

Dr Muir obtains the resultant in the form

$$\begin{vmatrix} A & H & G & A' & H' & G' \\ H & B & F & H' & B' & F' \\ G & F & C & G' & F' & C' \\ A' & H' & G' & A'' & H'' & G'' \\ H' & B' & F' & H'' & B'' & F'' \\ G' & F' & C' & G'' & F'' & C'' \end{vmatrix} = 0,$$

where A, B, C, F, G, H are the inverse coefficients of U , and similarly for V and W .

This is a perfectly correct result; but it is to be observed that the resultant is of the degree 12 in the coefficients of each of the functions U, V, W , and is therefore of the degree 36 in all. But the resultant of three quadric surfaces is known to be of the degree 24 only; and the foregoing form thus contains the resultant as a factor, multiplied by an extraneous factor of the degree 12.

The explanation is as follows: the condition that the three quadrics may have a common point is not that the determinant may vanish, but that the determinant and its first minors may all of them vanish; or, what is the same thing, that the matrix

$$\begin{pmatrix} A & H & G & A' & H' & G' \\ H & B & F & H' & B' & F' \\ G & F & C & G' & F' & C' \\ A' & H' & G' & A'' & H'' & G'' \\ H' & B' & F' & H'' & B'' & F'' \\ G' & F' & C' & G'' & F'' & C'' \end{pmatrix}$$

may be of rank 3.

But Dr Muir's result may be transformed as follows: writing

$$\begin{aligned} L &= -ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy, \\ M &= ax^2 - by^2 + cz^2 + 2fyz + 2gzx + 2hxy, \\ N &= ax^2 + by^2 - cz^2 + 2fyz + 2gzx + 2hxy, \end{aligned}$$

the equations $U = 0, V = 0, W = 0$ may be replaced by $L = 0, M = 0, N = 0$; and we have identically

$$\begin{aligned} L + fP &= x(Ax + Hy + Gz), \\ M + gQ &= y(Hx + By + Fz), \\ N + hR &= z(Gx + Fy + Cz), \end{aligned}$$

where P, Q, R are certain quadric functions. But L, M, N are sums of mere numerical multiples of U, V, W , viz. we have

$$\begin{aligned} 2L &= -aU + bV + cW, \\ 2M &= aU - bV + cW, \\ 2N &= aU + bV - cW, \end{aligned}$$

so that the original equations $U = 0, V = 0, W = 0$ are equivalent to and may be replaced by $L = 0, M = 0, N = 0$.

Muir shows that we have identically

$$\begin{aligned} L - fP &= x(Ax + Hy + Gz), \\ M - gQ &= y(Hx + By + Fz), \\ N - hR &= z(Gx + Fy + Cz), \end{aligned}$$

and hence the condition for a common point is obtained in the required form.